Copernicus Global Land Operations "Cryosphere and Water"

"CGLOPS-2"

Framework Service Contract N° 941119 (JRC)

ALGORITHM THEORETHICAL BASIS DOCUMENT

LAKE ICE EXTENT (LIE)

LECTION 250M CONTINENTAL EUROPE

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List of Acronyms

ATBD	The Algorithm Theoretical Basis Document
CEURO	Continental Europe
C-GLOPS2	Copernicus Global Land Operations Lot 2
ESA	European Space Agency
FCDR	Fundamental Climate Data Records
FMI	Finnish Meteorological Institute
FSC	Fractional Snow Cover
LIE	Lake Ice Extent
MODIS	Moderate Resolution Imaging Spectroradiometer
MSI	Multi-Spectral imaging Instrument
NIR	Near Infrared
OLCI	Ocean and Land Colour Instrument
OLI	Operational Land Imager
QAR	Quality Assessment Report
S2	Sentinel-2
S3	Sentinel 3
SCE	Snow Cover Extent
SLSTR	Sea and Land Surface Temperature Radiometer
Syke	Finnish Environment Institute
TIRS	Thermal Infrared Sensor
TOA	Top of atmosphere
VIIRS	Visible Infrared Imaging Radiometer Suite

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1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

Copernicus Global Land Operations Lot 2 (C-GLOPS2) is an operation of the Global Land Component, thematic domain Cryosphere and Water. It is allocated as a component of the Land service to ensure the ramp-up and the operational production of "Cryosphere and Water" related bio-geophysical parameters. Lake Ice Extent 250m is one of the Copernicus Global Land Service products. CEURO LIE 250m is a successor of the legacy product LIE BALTIC 250m.

The CEURO LIE 250m classifies, on a pixel basis, inland/freshwater bodies as (1) Fully snow-covered ice; (2) Partially snow-covered ice or snow-free ice; (3) Open water. A certain class is associated to a pixel following the rule of highest probability. Classification is provided only for cloud-free pixels, with dedicated cloud masks for the Lake Ice Extent product. The gridded data product covers Continental Europe [72°N / 11°W to 35°N / 50°E] in WGS-84 geographical coordinate system with a resolution of 0.0025 degree (~250m, depending on latitude).

This document describes the algorithm for the CEURO LIE 250m product.

1.2 SCOPE AND OBJECTIVES

One of the main objectives of Copernicus programme is to provide information on several environmental variables to the scientific community as well as other stakeholders including policy makers. This has been detailed in the Service Specifications Document (GIOGL1-ServiceSpecifications). The products are then operationally generated and delivered freely through the Copernicus Global Land portal (http://land.copernicus.vgt.vito.be) and through FTP distribution in near real-time as well as in offline.

This Algorithm Theoretical Basis Document (ATBD) describes the algorithm for the LIE 250m. The purpose of this document is to provide theoretical background for the CEURO LIE 250m satellite data product.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the users requirements, and the expected performance
- Chapter 3 describes the Lake Ice Extent (LIE) retrieval methodology
- Chapter 4 lists the relevant references

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Technical Specifications JRC/IPR/2015/H.5/0026/OC to Contract Notice 2015/S 151-277962 of 7th August 2015

AD2: Appendix 1 – Copernicus Global land Component Product and Service Detailed Technical requirements to Technical Annex to Contract Notice 2015/S 151-277962 of 7th August 2015

AD3: GCOS – 245 : The 2022 GCOS ECVs Requirements (GCOS 245), WMO, 2022

AD4: Copernicus Global Land Component Lots 1, 2 & 3 Operation of the Global Land Component (Framework Service Contract N° 199494, 199496, 199497, 941119): Service Specifications (CGLOPS-SSD). Issue 1.00. 24th November 2016.

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service.
CGLOPS2_PUM_LIE_250m-V2.2.1_I1.03	Product User Manuals summarizing all information about the LIE 250m product
CGLOPS2_QAR_LIE-250m-V2.2.1_I1.03	Report describing the results of the scientific quality assessment of the LIE-250m product

1.4.3 Output

Document ID	Descriptor
CGLOPS2_ATBD_LIE_250m-V2.2.1_I1.03	Algorithm Theoretical Basis Document of the LIE 250m product

2 REVIEW OF USERS REQUIREMENTS

According to the applicable document [AD2], the user's requirements relevant for European LIE 250m are:

- **Definition**: Continental Europe Lake Ice Extent (CEURO LIE)

- **Geometric properties**:
  - The target baseline location accuracy shall be 1/3 of the at-nadir instantaneous field of view.
  - Pixel coordinates shall be given for centre of pixel.

- **Geographical coverage**:
  - Projection: geographical latitude, longitude
  - Geodetical datum: WGS84, EPSG:4326
  - Pixel size: 0.00025° (ca. 250 m, depending on latitude)
  - Coordinate position: pixel centre
    - Window coordinates:
      - Upper left: 11ºW, 72ºN
      - Lower right: 50ºE, 35ºN

- **Accuracy requirements:**

  Target requirements for lake ice cover based on Global Climate Observing System (GCOS) (AD4) are following:

  - Spatial resolution: 300 m
  - Temporal resolution: daily
  - Measurement uncertainty: ≤10%

User requirements related to LIE products are described by the GCOS – 245 : The 2022 GCOS ECVs Requirements (GCOS 245) (AD3) and in User Requirements of CryoLand Snow and Land Ice services – Cryoland (Malnes et al. 2015).

The main user groups interested in these satellite products are the climate and hydrological modelling communities and weather forecasting authorities. The following paragraphs will summarize the thematic, temporal, and geometric aspects concerning the LIE 250m product.

Thematic aspects

The LIE 250m product classifies a section of the freshwater body as (1) Fully snow-covered ice, (2) Partially snow-covered ice/clear ice, and (3) Open water. Additionally, a class is given for fully snow-covered ice. The targeted minimum measurement uncertainty is 10% according to AD3.

Geometric aspects

Mapping the ice on lakes over the Continental Europe with a daily coverage requires at the moment the use of moderate resolution imagery due to the lack of daily high-resolution data. However, mapping of ice on small lakes requires a fine spatial resolution. The current spatial resolution for the parameter LIE is 250 m for optical data, with a geolocation accuracy of 1 pixel (Fernández-Prieto et al. 2012). NOAA-20 (JPSS-1) VIIRS (Visible Infrared Imaging Radiometer Suite) data provide good daily coverage over Continental Europe, with multiple overlapping overpasses per day.

Temporal aspects

The LIE 250 m is a daily product. Spatially and temporally consistent data sets are important for climate modelling and numerical weather prediction. Climate modellers need data sets over large areas (continental scale) and for long time periods (decades). Aggregated (gridded) data sets are preferred. The data sets do not need to be very frequent in time (monthly information is sufficient according to Climate modellers). For other than climatic use, Malnes et al. (2015) summarize that the majority of users request a one day temporal resolution. When often bridging the wishful thinking from users with what is actually possible to achieve with current and near future satellite sensors, it should be realistic to obtain a temporal resolution of one week, due to frequent cloud cover at high latitudes where lake ice mostly occurs. There is hence a need to also provide a level of refinement to the ice products, where temporal interpolations provide the desired temporal regularity.

Near real-time data:

The climate modellers have little need for real-time data: annual updates (mostly concerning ice break-up) are sufficient.

Weather forecasters may in principle need near real time data, or at least data for the latest date. The most urgent need for real time data seems to arise from within the hydrological community, for whom the real time data is a pre-requisite for flood management.

Other aspects

Satellite data on freezing and break-up dates of lake ice is also of interest to all approached users, particularly for the validation of models. These data types can also be of interest for environmental assessment and for biological monitoring.

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3 METHODOLOGY DESCRIPTION

3.1 OVERVIEW

The Continental Europe Lake Ice Extent (CEURO LIE) 250m resolution product of the Copernicus Global Land Service provides three alternative classes for each pixel of a water body: (1) Fully snow-covered ice, (2) Partially snow-covered ice/clear ice, and (3) Open water. This classification may be performed from optical satellite sensors of different resolutions. The classification is available for cloud-free pixels with a dedicated cloud mask. The main use of the data is to define timing of events in the annual cycle of freshwater ice, i.e., the initiation of ice formation, total freeze up of lakes, initiation of melting period, and ice-out date (the day when the lake is completely ice-free). The gridded product can also be used to monitor the extent of ice, and this is usually defined in terms of area (in square kilometers) or percentage of coverage.

3.2 THE RETRIEVAL ALGORITHM

3.2.1 Outline

The processing chain of LIE is displayed in

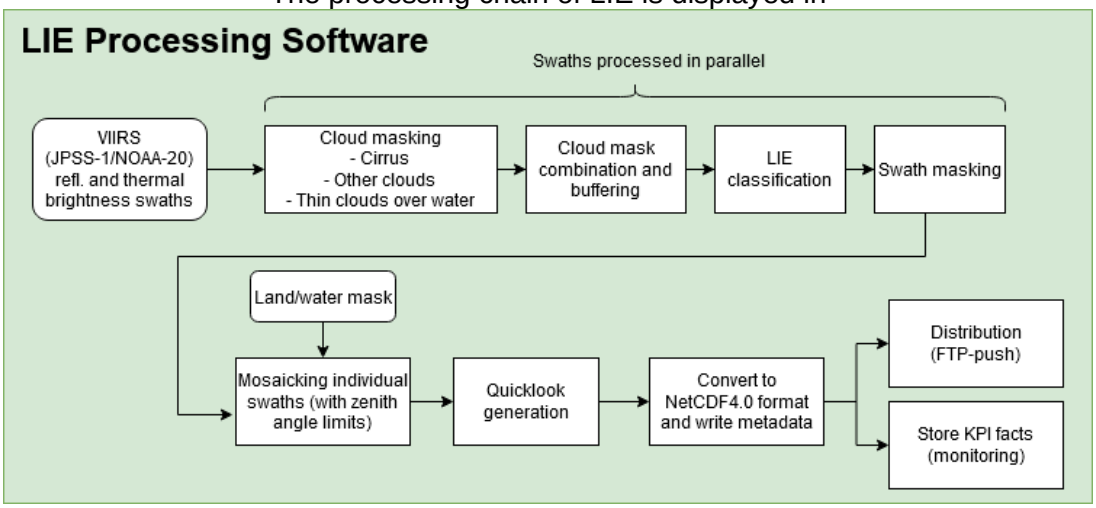


Figure 1. The chain takes in pre-processed top-of-atmosphere (TOA) reflectance and thermal brightness temperature data. Near Infrared (NIR, ~865nm) reflectance thresholds are utilized to determine the LIE classes. The LIE estimate, cloud masks and lake/sea/land masks are combined

to create products from individual swaths (there are 2-4 swaths used per day). Individual swaths are mosaicked, rejecting data from sun elevation < 15°.

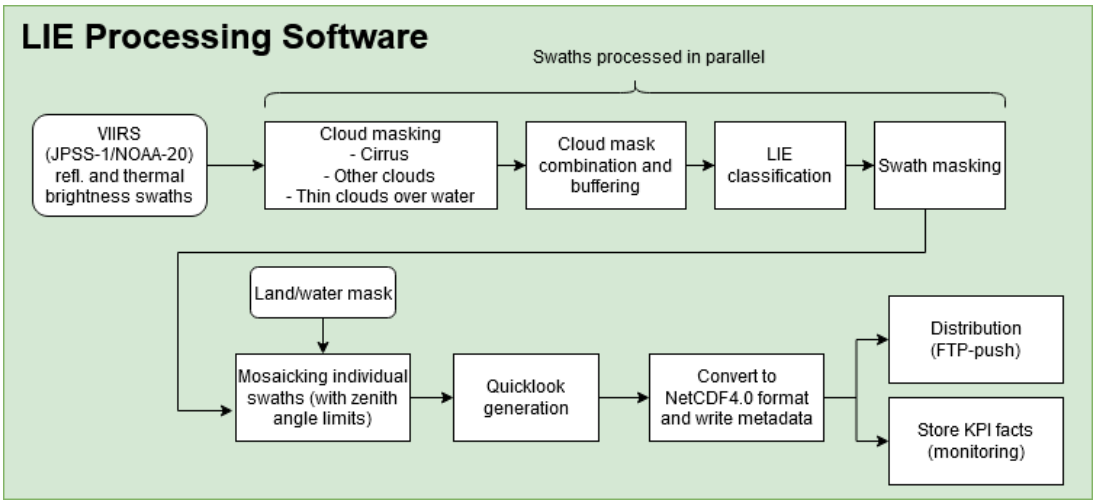


Figure 1. Processing chain for LIE.

3.2.2 Basic underlying assumptions

- Snow, (clear) ice and open water can be clearly distinguished from reflectance in the near infrared (NIR, ~865 nm) wavelengths (Latifovic and Pouliot 2007; Perovich 1996)
- Water properties, i.e., the concentration of clorophyll-*a* and suspended matter in water cause a mix-up between clear ice and open water only when these optically active substances reach very high values (Doxaran et al. 2002; Ritchie et al. 2003)
- The algorithm uses the Top-of-Atmosphere (TOA) reflectance. The errors due to atmosphere and viewing geometry are considered acceptable due to the simplicity of the algorithm. Also, limitations of usable viewing and sun zenith angles have been set up. The effects of topography in the reflectance can be neglected in case of lake ice.
- The abovementioned errors are also small in the ice phenology calculation, where the lack of observations owing to cloud cover and low light conditions are causing a considerably higher error.

3.2.3 Related and previous applications

The Copernicus LIE 250m product and service builds on the legacy of the pre-operational CryoLand LIE product and service for Europe, developed and implemented within the EU FP7 project (EU FP7 No 262925). The algorithm for ice classification was developed at Syke. The original algorithm at Syke, used for monitoring the spring-time ice conditions in Finnish inland waters (snow cover on lake ice), was a direct application of the Fractional Snow Covered area (FSC) -algorithm, which was using Terra/MODIS channel 1 (620–670 nm) data (Metsämäki et al. 2012; Metsämäki et al. 2005). Along with the service running, it was found that the distinction between the melting stages of lake ice and overlying snow cover is very challenging when using medium resolution optical images. As the main application areas of the earlier lake ice product was to follow the spring break-up of the lake ice cover and to potentially use of the product in numerical weather simulations, the snow cover on ice was considered less important and the emphasis was put on the ice extent. As a consequence, a new algorithm for lake ice mapping was developed. In addition to omitting the snow cover fraction over the ice, the deployed band was changed from MODIS channel 1 to channel 2 (841–876 nm), since channel 2 is more sensitive to changes between ice/snow and water, and it is not so sensitive to changes in the water quality, when turbidity remains relatively low. The new product was given three ice classes and water class: "Snow-covered ice", "Partial snow/white ice", "Clear ice", "Open water". The Copernicus LIE 250m - classification is made more robust by reducing the classes to (1) Fully snow-covered ice; (2) Partially snow-covered ice/clear ice; (3) Open water. This is justified by the fact that clear ice and partially snow-covered ice/white ice do not provide sufficiently different spectral signal for reliable separation. Therefore, the product has stronger focus now on the extent of ice on lakes. The fully snow-covered ice still remains a separate class. Later, thermal channel was employed to minimizing misinterpretations during the summer, such as those caused by phytoplankton blooms. When the sensor was transferred from MODIS to VIIRS, the area was extended from Baltic region to cover Continental Europe and the nearest VIIRS equivalent bands were used with respect to the MODIS observation wavelengths. Compared to MODIS Lake Ice Extent dataset (2017–), the updated VIIRS-based CEURO LIE offers a more advanced cloud algorithm to enhance cloud screening, including thin clouds over cold water.

3.2.4 Input data

The algorithm uses top-of-atmosphere (TOA) data from optical satellite instruments. The LIE 250m classifier exploits NIR reflectance (~865 nm) and thermal band (~11.5 µm). Other bands are used

for cloud masking (488 nm, 555 nm, 865 nm, 1.4 µm, 1.6 µm, 3.7 µm, and 10.8 µm). The CEURO LIE 250m product relies on NOAA-20 VIIRS 375m resolution data. However, because the simplicity of the algorithm, it is easily transferred to any other optical satellite instrument. The algorithm benefits from the spatial averaging of medium resolution satellites (spatial resolution in the order of 100 m), but it is also applicable to higher -resolution instruments (spatial resolution of the order of 10–20 m).

Following satellite instruments are also possible for producing LIE:

- Terra/MODIS: An instrument employed in the operative Copernicus Baltic LIE product between 2017 and 2024.
- Sentinel-3 Ocean and Land Colour Instrument (OLCI)
- Sentinel-3 Sea and Land Surface Temperature Radiometer (SLSTR): An instrument employed in the operative Copernicus Northern Hemisphere LIE 500m resolution product since 2021.
- Sentinel-2 Multi-Spectral imaging Instrument (MSI): Can be used for validation of medium resolution data, as well as for producing higher resolution ice cover.
- Landsat-8/9 Operational Land Imager (OLI)/ Thermal Infrared Sensor (TIRS): Can be used for validation of medium resolution data, as well as for producing higher resolution ice cover.
- Suomi-NPP and NOAA-21 VIIRS: Could be used as back-up data source. Of these, NOAA-21 is more modern and could be regarded as the primary choice.

3.2.4.1 Auxiliary data

Land areas are masked for better visualization and for exclusion of mixed (land/water) pixels. The land mask needs to be as accurate as possible and buffered to avoid misinterpretations for shoreline pixels and for shallow waters with vegetation. Especially small-featured water bodies, such as the lakes in Finland, require accurate masking. The current land/sea/lake mask is based on following datasets:

- European Commission Global Surface Water: Water Occurrence Data
- Global Self-consistent, Hierarchical, High-resolution Geography Database (GSHHG)

Land and sea areas are masked out from the final product.

3.2.5 Output product

The CEURO LIE 250m product is provided on a daily basis in NetCDF format. A quicklook file in GeoTIFF format (1km resolution) is generated for each daily LIE product. Figure 2 presents the samples of the data.

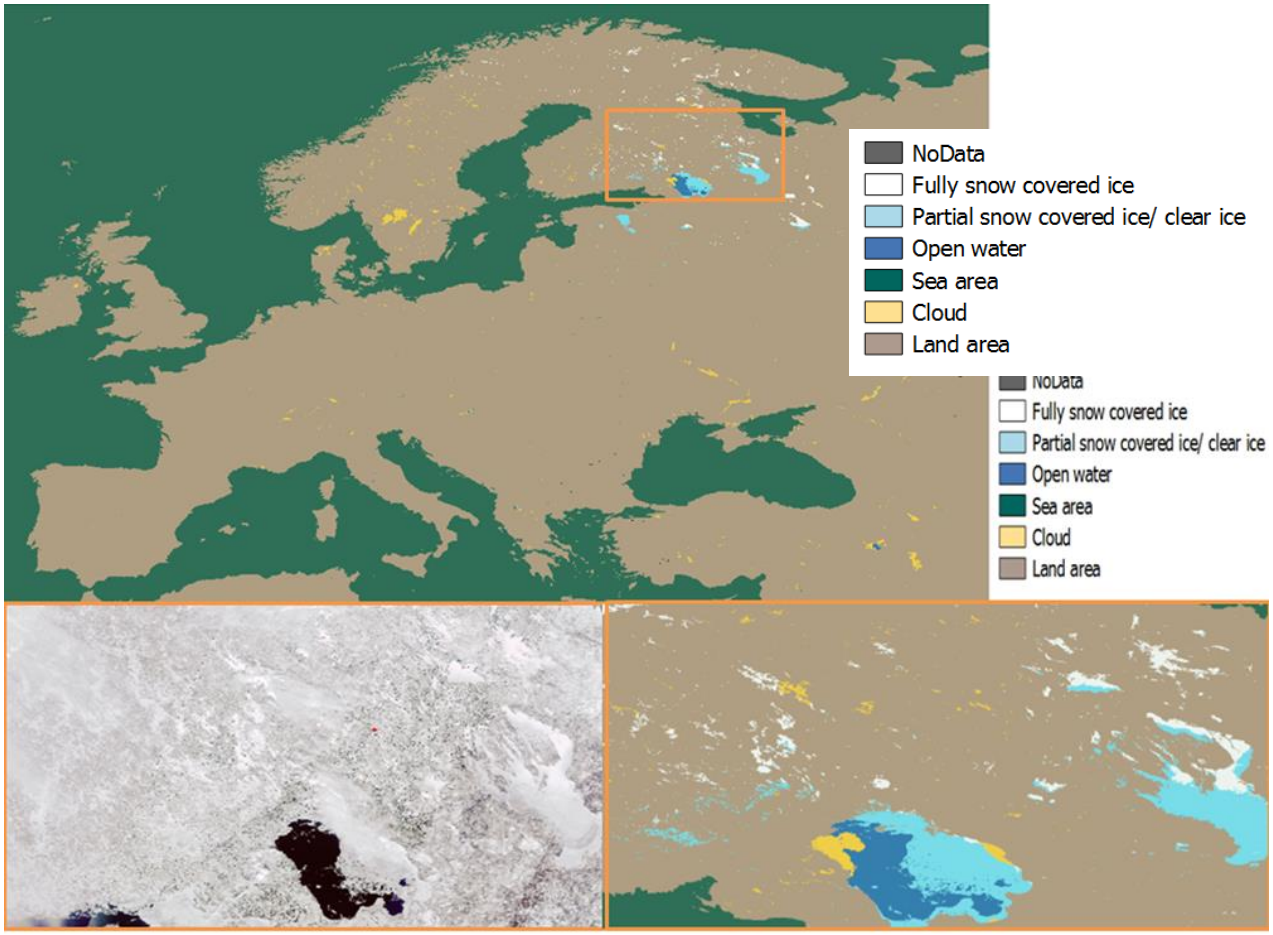


Figure 2. CEURO LIE 250m product (17 March 2023). *Top:* The whole Continental Europe Lake Ice Extent product, *Bottom:* a closeup to Finland and its surroundings.

3.2.6 Methodology

Algorithm is based on reflectance thresholds determined by comparing time series of satellite reflectances to the ice thickness observations over selected monitoring sites (a selection from sites shown in Figure 3) and observations of ice break-up. It has been shown that there are clear differences in the reflectance response of snow, clear ice, and open water (Latifovic and Pouliot

2007; Perovich 1996). The algorithm is based on these differences. This is relatively simple, as there are large differences in the reflective properties of ice and water in the visible and near-infrared spectrum (Perovich 1996). There is also clear difference in the reflectance of clear ice and snow cover. Therefore, the distinction of lake ice with full snow cover is also relatively straight forward. This can be useful information if the data would be used as reference for energy balance calculations in thermodynamic models of lake ice. What lie in between full snow cover and clear ice are patches of snow over clear ice or snow-ice (flooded and re-frozen snow) in different composition. The clear ice class also includes the mixture class of ice and open water, i.e. areas where the conditions allow the ice floes to remain close to each other, yet retaining some open water in between. Once the ice is in this state the effects of wind and radiation easily break up the rest of the ice making it melt quickly. Therefore, these conditions do not last long periods and the water quickly becomes ice free. Classifying these areas as open water is therefore acceptable. The NIR-channel for ice detection was selected due to its insensitivity to variations in other water properties (chlorophyll content, suspended matter etc.) when turbidity is low (Doxaran et al. 2002; Ritchie et al. 2003). The reflectance is dependent on the solar incidence angle and viewing angle of the satellite, but when developing the LIE algorithm, these effects were overlooked and the reference reflectances were optimized with MODIS reflectance time-series from 300 arbitrary pure water pixels (match-ups) from lakes with in situ measurements in Finland in year 2011.

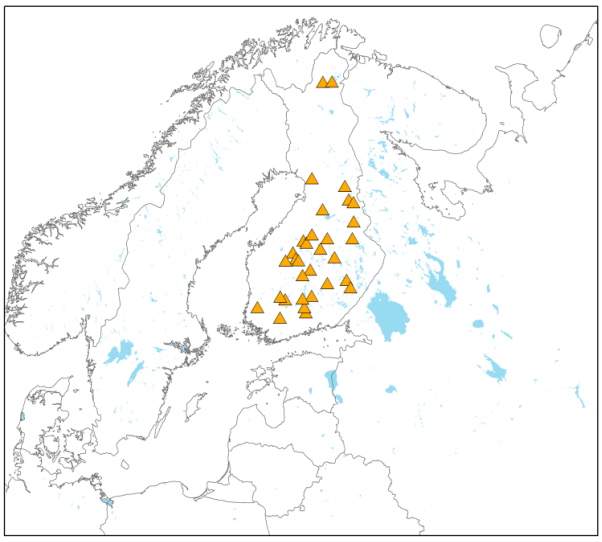


Figure 3. Map of *in situ* observation sites of ice phenology in Finland

In the determination of the reference reflectances, the extensive in-situ monitoring network of Syke was of great support (Figure 3). Syke monitors the ice coverage, ice thickness and thickness of snow

cover over lake ice at several sites in Finnish inland waters. Finding the reference reflectances for the algorithm was based on comparing time series of MODIS/Terra reflectances to the snow thickness observations over selected monitoring sites. The temporal change of the reflectance over a lake follows the description shown in Figure 4. The different surface classes on a lake (1) Fully snow-covered ice; (2) Partially snow-covered ice/clear ice; (3) Open water can be clearly discriminated using reflectance values from optical satellite data. In addition, thermal data is utilized to eliminate possible misinterpretations during the summer, e.g., caused by a phytoplankton bloom event. While transferring the sensor from MODIS to VIIRS, the reflectance thresholds were adjusted for VIIRS bands by compering reflectances with hundreds of high-resolution images where ice conditions were visible.

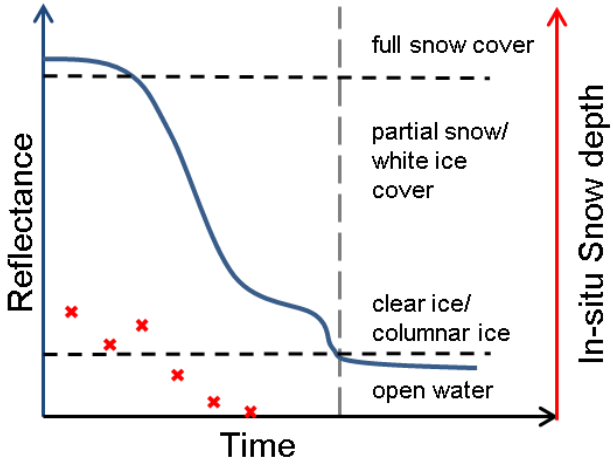


Figure 4. Schematic illustration of reflectance time-series from satellite instrument (blue line), measurements of snow thickness on ice (red crosses) and in-situ observation of ice disappearance.

3.2.7 Limitations

Continental Europe Lake Ice Extent (CEURO LIE) is monitored daily using optical satellite data. Observations from optical satellite sensors are restricted by cloud cover as well as low light conditions from late autumn to early spring in the high northern latitudes due to the polar night.

Mapping of ice on lakes requires, especially in Fennoscandia, fine spatial resolution watermask due to the small size of the lakes and relatively high number of islands and narrow parts in some lakes. The gridded CEURO LIE data product covers the Continental Europe (see geographical coverage above) with 0.0025° (~250m) resolution. The shores (also in case of island) are not included in the

product to avoid mixed pixels, i.e., containing both land and water/ice). The user should be aware that during the freezing and break-up of ice the method cannot always detect whether the target represents water on dark (thin, semitransparent) ice or open water. Same goes for fact that the small white ice floes within a single pixel may results the pixel being classified as ice. However, in these cases the lake ice is already fragmented and/or dark and weak and thus does not always cause disadvantage for the end users depending on the user case. Currently the processing includes all water bodies (lakes and large rivers), which can be distinguished by the auxiliary data (see section 3.2.4.1).

3.3 QUALITY ASSESSMENT

The algorithm is based on reflectance thresholds determined by comparing time series of MODIS reflectances to the snow thickness observations over selected monitoring sites and observations of ice break-up. Syke has unique in-situ datasets of lake ice observations dating back to the end of 19th century. The observations are still operational. Table 1 lists the datasets available for validation. When switching the sensor from MODIS to VIIRS, the reflectance thresholds were adjusted for VIIRS bands by compering reflectances with hundreds of high-resolution images where ice conditions were visible.

Table 1. Validation datasets

In-situ dataset	*Freezing and breakup*	*Ice thickness and snow on ice thickness*
Unit	Date	cm
Accuracy	±1 day	±1 cm
Observation interval	Daily observations	3 per month (10th, 20th and 30th)
Areal extent	Entire observation period: 122 sites	Entire observation period: 161 sites
Temporal coverage	1753–2015 (changing network)	1912–2015 (changing network)

In future, the Lake Ice Extent 250m product will be spatially extensively validated against high resolution LIE products derived from optical satellite data of Sentinel-2, with Landsat-8 as back-up applying procedures and protocols developed in the EU FP7 project CryoLand, including statistical information.

3.4 RISK OF FAILURE AND MITIGATION MEASURES

Sensor failure

Sensor failure is always a risk in remote sensing projects. LIE 250m algorithm utilize currently NIR and thermal channels of NOAA-20 VIIRS data. Suomi-NPP VIIRS data will be used as back-up input data. The processing system for this sensor is already implemented at FMI. The possibility of both missions' failure is considered to be small. The LIE method is also adaptable to Sentinel-3 OLCI (for ice coverage) & SLSTR (for ice and cloud detection) sensors. The probability for complete sensor failure is low for S-3, as the sensor is already in orbit, and provides data within the commissioning.

Low storage capacity

The amount of raw data delivered daily by the NOAA-20 VIIRS sensor for the Continental Europe is large, and storing data is challenging. Thus, there is a risk of limited storage capacity. This risk is expected to be low, as FMI is prepared to store large amounts of data and for the production, NOAA-20 VIIRS data are fetched from the NOAA Comprehensive Large-Array Data Stewardship System (CLASS) and are deleted after successful product generation to save disk space.

Timely Satellite Data availability

In order to provide the product in Near Real Time (latency time < 3 days), satellite data need to be available within 3 hours after image acquisition.

Processing Risk

Continental Europe LIE 250m product generation is a computationally intensive task, especially due to the large amount of input data to be processed daily. There is a certain risk in processing this large amount of data in a short time, depending on the availability of computation power at FMI. As the software handling large amount of data is recently improved, we rate this risk as low.

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